Brief Announcement: Strong and Hiding Distributed Certification of k-Coloring

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Abstract

We study the problem of certifying whether a graph is k-colorable with a locally checkable proof (LCP) that is able to *hide* the k-coloring from the verifier, in the sense that no algorithm can (completely) extract a k-coloring from the certificate. Motivated by the search for promise-free separations of extensions of the LOCAL model in the context of locally checkable labeling (LCL) problems, we also require the LCPs to satisfy what we call the *strong soundness* property. We focus on the case of 2-coloring and show that strong and hiding LCPs for 2-coloring exist in specific graph classes and require only $O(\log n)$-sized certificates. Furthermore, when the input is promised to be a cycle or contains a node of degree 1, we show the existence of strong and hiding LCPs even in an *anonymous* network and with *constant-size* certificates. Despite these upper bounds, we prove that there are no strong and hiding LCPs for 2-coloring in general, *regardless of certificate size*. Along the way, we give a characterization of the hiding property for the general k-coloring problem that appears to be a key component for future investigations in this context.

CCS Concepts

• **Theory of computation → Distributed algorithms**.

Keywords

coloring problem, distributed certification

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1 Introduction

Distributed certification is a framework in distributed computing where nodes in a network collaboratively verify that the entire network (graph) satisfies a given property. Within this framework, a *locally checkable proof (LCP)* is a non-deterministic distributed algorithm designed to verify global properties of a graph G. It involves two components: a *prover* and a *distributed verifier*. The prover is an omnipotent computational entity, capable of performing any Turing-computable operation instantaneously. Its role is to convince the distributed verifier, which consists of a collection of nodes within the graph, that G satisfies a particular property $\mathcal{P}$. The LCP must satisfy the properties of *completeness* and *soundness*, which are defined in a manner akin to the classical proof systems from computational complexity theory.

In this paper,[1] we focus on *k-coloring*, which consists of determining whether the input graph is k-colorable. This problem admits a simple one-round LCP with certificates consisting of $O(\log k)$ bits, representing the color of the node in a proper k-coloring. Our goal is to find a way to certify k-coloring *without revealing* a proper k-coloring to the nodes. To achieve this, we introduce the notion of *hiding* locally checkable proofs. Informally, an LCP for k-coloring is said to be hiding if there exist certificates that convince the verifier on every k-colorable graph, but there is no local algorithm that can extract a k-coloring (of all nodes) from these certificates.

In addition to the hiding property, we also require our LCPs to satisfy a stronger but still very natural variant of soundness, which we accordingly dub *strong soundness*. The main reason for using this instead of the usual one is a connection to *promise-free separations of extensions of the LOCAL model*, which we discuss next.

Main motivation: application to promise-free separations. *Locally checkable labeling* (LCL) problems are central to the study of the LOCAL model of computation [22, 24]. These are problems where one must label the nodes of a graph according to constraints that can be locally verified. By far, the most studied variant of LOCAL is randomized LOCAL [4, 5, 9, 11, 13, 15, 17, 25], but recently interest has extended to several other augmentations of the LOCAL model [1], including access to shared randomness [6], quantum computing [3, 14, 21], and global symmetry breaking [2, 12, 17, 18].

Seeing we are confronted with a landscape of several different models, we would like to prove separation results to ensure they are all indeed distinct. In particular, we would like to obtain *promise-free* separations that do not impose additional assumptions on the input graph. Nevertheless, proving such separations is usually an ordeal and only *ad hoc* constructions are known (see, e.g., [6]).

For concreteness, let us focus on the case of separating the SLOCAL and online LOCAL models.[2] There we have an exponential separation for the 3-coloring problem, but only when restricted to bipartite graphs. More precisely, there is an $O(\log n)$ algorithm for 3-coloring bipartite graphs in online LOCAL, whereas in SLOCAL we have a polynomial lower bound that holds even in grids [2].

[1] See [23] for the full version.
[2] We do not require familiarity with the models for this high-level discussion. The interested reader is referred to, e.g., [1, 2].



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One simple approach to turn this into a promise-free separation would be to encode in the LCL input a certificate of 2-colorability while asking for a 3-coloring (i.e., the nodes must label themselves with a proper 3-coloring). Since we must allow arbitrary input graphs, the certificate cannot be valid in all of them, so we require the nodes to output a correct 3-coloring *only on the parts of the graph where the certificate is valid*. In addition, for it to be useful, the certificate must be *hiding* in the sense that every SLOCAL algorithm with $O(\log n)$ locality fails to extract a 3-coloring from it.

Let us refer to this LCL problem, that is, 3-coloring under the presence of a certificate of 2-colorability as Π. Assuming such a hiding certificate exists, this yields a *promise-free separation* between SLOCAL and online LOCAL: SLOCAL fails to solve Π (with $O(\log n)$ locality) since it cannot extract a 3-coloring from the certificate, while online LOCAL can simply execute the existing $O(\log n)$ coloring algorithm on the parts of the graph where the certificate is valid (and labels the rest of the nodes arbitrarily).

Alas, the only way of certifying bipartiteness that is known is to reveal a 2-coloring, which renders the 3-coloring problem trivial. Hence it would be amazing to have a certificate for bipartiteness that *hides* a 3-coloring. In addition, this certificate must be resilient against being used to label graphs that are not bipartite. That is, for any graph G where someone tries to place the certificate labels on, the parts of G where the certificate is valid is indeed 2-colorable (so that the online LOCAL algorithm succeeds in 3-coloring these parts of G). This motivates the stronger variant of soundness that we introduce. We discuss these two notions in more detail next.

2 The model

As mentioned above, the goal of the prover in the LCP model is to convince the distributed verifier that the underlying graph G satisfies a certain property $\mathcal{P}$. To achieve this, the prover assigns a *certificate* $\ell(v)$ to each node v. Once these certificates are distributed, each node v receives the certificates of its vicinity. Using this local information—its own certificate, the certificates of its neighbors, and the graph structure visible to it—each node independently decides whether to accept or reject the assertion that G satisfies $\mathcal{P}$.

An LCP is *correct* if it satisfies *completeness* and *soundness*. Soundness guarantees that if all nodes accept the assigned certificates (i.e., the proof is locally verified as correct at each node), then the underlying structure genuinely satisfies the intended property. Completeness, on the other hand, ensures that whenever the structure does satisfy the intended property, there is indeed a way to assign certificates such that local verification at every node passes.

In the standard variant of soundness, if G does not satisfy $\mathcal{P}$, then, for any assignment of certificates that a malicious prover attempts to distribute, there will be at least one node in G that rejects. Here we impose the following strengthening: If G does not satisfy $\mathcal{P}$, then, for any assignment of certificates, *the subgraph induced by accepting nodes of* G must satisfy the property $\mathcal{P}$. This is an equally natural requirement, besides being the correct notion for the aforementioned connection to LCL problems.

We consider LCPs where the verifier runs for r rounds, for a fixed positive integer r, which we refer to as *r-round* LCPs. Hence every node "sees" the graph structure and all certificates up to r hops away. Every node is also given a unique identifier smaller than a value N that scales polynomially with the number of nodes and is known by the nodes of the graph. If the LCP operates independently of the values of the identifiers, we refer to it as an *anonymous* LCP.

Hiding LCPs. The central novel notion of our interest is that of a *hiding* LCP. Focusing on the case of coloring, an r-round LCP for k-coloring is hiding if no r-round local algorithm can extract a k-coloring from a correct certificate (i.e., a certificate that is accepted unanimously by the verifier). Note that "failing to extract" means it is acceptable if a *single* node fails to output its color. Hence this is a purely algorithmic notion and is far from having cryptographical applications or similar. Also notice that not every graph property admits a hiding LCP on every graph. This contrasts with classical LCPs, where it is known that every Turing-computable graph property $\mathcal{P}$ admits an LCP with certificates of size $O(n^2)$.

3 Similar concepts and related work

The notion of hiding LCPs is reminiscent of zero-knowledge proof systems, which have also been studied in the distributed setting [7, 20]. Note zero knowledge is a stronger notion usually with applications of a cryptographical nature and requires the verifier to essentially learn nothing about the solution. The hiding property is more lenient in that it is sufficient for the solution to not be revealed everywhere. In the case of coloring, for instance, it is enough if a single node fails to output a correct color. This connects better to possible applications to LCL problems, as previously discussed.

Still in a complexity-theoretical vein, the question of whether there is a hiding LCP for a problem Π relates to whether the *decision* and *search* versions of Π are equal in complexity or not in the context of distributed proof systems. More specifically, a (non-adaptive, locally computable) reduction from the search to the decision version of Π contradicts the existence of a hiding LCP for Π. This is because what such a reduction is doing is, indeed, extracting the solution for Π from a correct certificate assignment.

A related model is the *resilient labeling scheme* [10, 16], a type of proof system where the certificates of up to f nodes can be erased. Unlike ours, this model imposes a requirement on *completeness* rather than *soundness*: In *yes-instances*, all nodes must accept the verification, even when a subset of nodes has lost their certificates.

There are several articles that deal with lower bounds related to k-coloring, beyond the trivial upper bound of $\lceil \log k \rceil$ bits (i.e. give each node its color in a proper k-coloring). See, for instance, [8, 19]. In this work, we ask whether any distributed proof that a graph is k-colorable *necessarily leaks* a proper k-coloring of the graph. This question may be related to the minimum certificate size required to prove that a graph is k-colorable, but it is not the same. Indeed, it is possible that in order to achieve strong and hiding LCPs for k-coloring, we need certificates that are strictly larger than those required to certify the property without any additional conditions.

4 Main results

Our main results are focused on the 2-coloring problem. Nevertheless, our framework for lower bounds is also applicable to k-coloring for arbitrary values of k. Some of our upper bound techniques are also useful in the general case, but we do not address those since the 2-coloring case is the most fundamental.

Upper bound with constant-size certificates. Our first result is the cornerstone case where the LCP is anonymous and we wish to certify a 2-coloring using only constant-size certificates. We prove it is possible to certify the 2-coloring without revealing it provided that we are in graphs with minimum degree one or even cycles (or a union of both). In the statement below, 2-col$_{\mathcal{H}}$ refers to the promise version of the 2-coloring problem where we allow only input graphs from the class $\mathcal{H}$.

Theorem 4.1. *Let $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2$, where $\mathcal{H}_1$ is the class of graphs G for which $\delta(G) = 1$ and $\mathcal{H}_2$ is the class of even cycles. Then there is a strong and hiding one-round anonymous LCP for* 2-col$_{\mathcal{H}}$ *using constant-size certificates.*

We use two distinct strategies for the classes $\mathcal{H}_1$ and $\mathcal{H}_2$. If the input graph has minimum degree one, we can reveal the coloring everywhere except for a node with degree one of our choosing. Meanwhile, if we are in a cycle, we can use a 2-*edge*-coloring, which hides the 2-coloring *from all nodes*.

Lower bound. We prove a matching lower bound for the case $r = 1$. When r is larger, our impossibility result holds for graphs that satisfy a certain connectivity property, which we define next.

Let r be a positive integer. A graph G is said to be *r-forgetful* if, for every node $v \in V(G)$ and every neighbor u of v, there is a path $P = (v_0 = v, v_1, \ldots, v_r)$ of length r such that, for every $w \in N^r(u)$, $\mathrm{dist}(v_i, w)$ is monotonically increasing with i. The intuition is that, if we are visiting nodes in G and arrive at v from u, then we can always find a way of "escaping" v without having to backtrack and go through the r-neighborhood of u. This property applies to a broad class of graphs, including (regular) grids and trees. A necessary condition is that the graph has diameter at least $2r + 1$.

Our main result is that it is impossible for r-round LCPs to hide a 2-coloring in r-forgetful graphs except for the two cases of Theorem 4.1. This holds even if the LCP is not anonymous.

Theorem 4.2. *Let r be a positive integer and $\mathcal{H}$ a class of graphs containing at least one r-forgetful graph that is neither an even cycle nor has minimum degree equal to one. Then there is no strong and hiding r-round LCP for* 2-col$_{\mathcal{H}}$ *(no matter the certificate size).*

The proof of this result uses our main technical contribution, which is the characterization of a hiding LCP for k-coloring based on the *neighborhood graph of accepting local views*: An LCP for k-coloring is hiding if and only if the neighborhood graph on accepting views is itself *not* k-colorable. Besides the lower bound proof, this characterization is useful for proving the hiding property of our upper bound LCPs as well.

Upper bounds with variable-size certificates. Finally, we provide evidence showing that our impossibility result cannot be (fully) extended to the non-anonymous case with super-constant certificate size. More precisely, we define two graph classes, namely graphs that admit a *shatter point* and *watermelon graphs*, both of which admit strong and hiding LCPs for 2-coloring.

Say a graph G *admits a shatter point* if there is a node v such that $G - N[v]$ is disconnected (i.e., it has at least two distinct connected components). For this class, we obtain the following theorem:

Theorem 4.3. *Let $\mathcal{H}$ be the class of graphs G admitting a shatter point. Then, there is a strong and hiding one-round LCP for* 2-col$_{\mathcal{H}}$ *using certificates of size $O(\min\{\Delta^2, n\} + \log n)$, where $\Delta = \Delta(G)$ is the maximum degree of G.*

Meanwhile, a *watermelon graph* is a graph G defined by two nodes $v_1, v_2 \in V(G)$ and a collection of disjoint paths $\mathcal{P} = P_1, \ldots, P_k$ of length at least 2 with endpoints in v_1 and v_2. Nodes v_1 and v_2 are called the *endpoints* of G. In this case, we show a strong and hiding LCP for 2-coloring using certificates of size $O(\log n)$.

Theorem 4.4. *The class $\mathcal{H}$ of watermelon graphs admits a strong and hiding* 1*-round LCP for* 2-col$_{\mathcal{H}}$ *with $O(\log n)$-size certificates.*

We conclude with a discussion about our proof of Theorem 4.2. The key realization is that the hiding property can be fully characterized in terms of the *neighborhood graph* of yes-instances of an LCP. For an r-round LCP $\mathcal{D}$, we denote the respective neighborhood graph for instances with at most n nodes by $\mathcal{V}(\mathcal{D}, n)$. The characterization we prove is that, for any positive integer k, $\mathcal{D}$ hides a k-coloring if and only if $\mathcal{V}(\mathcal{D}, n)$ is *not* k-colorable.

The proof of the lemma is by showing how we can use a k-colorable $\mathcal{V}(\mathcal{D}, n)$ to extract a coloring for $\mathcal{D}$. The characterization suggests using non-k-colorable subgraphs of $\mathcal{V}(\mathcal{D}, n)$ in order to prove Theorem 4.2. That is, assuming that $\mathcal{D}$ hides 2-colorings, we would like to take a non-bipartite subgraph H in $\mathcal{V}(\mathcal{D}, n)$ and turn it into a real instance. Since we are in the special case of 2-colorability, it suffices if H is an odd cycle—or, more generally, an odd closed walk—in $\mathcal{V}(\mathcal{D}, n)$.

The main challenge is that, although $\mathcal{V}(\mathcal{D}, n)$ consists of views arising from accepting instances, there is no guarantee that we can form a real graph from such an arbitrary H. This problem is evident if H is a closed walk but not a cycle, but even then H may contain views that are inherently incompatible. For instance, distinct views of H may contain a node with the same identifier but a different number of neighbors. Indeed, if H can be realized directly, then it is *immediate* that $\mathcal{D}$ violates strong soundness.

We introduce and develop a notion of *realizability* of subgraphs of $\mathcal{V}(\mathcal{D}, n)$ with which we gradually manipulate an odd cycle of $\mathcal{V}(\mathcal{D}, n)$ into a real instance. The manipulations needed are fairly technical since we must ensure the resulting views all agree with each other. The question of realizability of subgraphs of the neighborhood graph is an interesting blend of graph theory and distributed computing that may be of independent interest.

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References

[1] Amirreza Akbari, Xavier Coiteux-Roy, Francesco D'Amore, François Le Gall, Henrik Lievonen, Darya Melnyk, Augusto Modanese, Shreyas Pai, Marc-Olivier Renou, Václav Rozhoň, and Jukka Suomela. 2025. Online Locality Meets Distributed Quantum Computing. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing, STOC 2025, Prague, Czech Republic, June 23-27, 2025*. ACM. arXiv:2403.01903

[2] Amirreza Akbari, Navid Eslami, Henrik Lievonen, Darya Melnyk, Joona Särkijärvi, and Jukka Suomela. 2023. Locality in Online, Dynamic, Sequential, and Distributed Graph Algorithms. In *50th International Colloquium on Automata, Languages, and Programming, ICALP 2023, July 10-14, 2023, Paderborn, Germany (LIPIcs, Vol. 261)*, Kousha Etessami, Uriel Feige, and Gabriele Puppis (Eds.). Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 10:1–10:20. doi:10.4230/LIPICS.ICALP.2023.10

[3] Alkida Balliu, Sebastian Brandt, Xavier Coiteux-Roy, Francesco d'Amore, Massimo Equi, François Le Gall, Henrik Lievonen, Augusto Modanese, Dennis Olivetti, Marc-Olivier Renou, Jukka Suomela, Lucas Tendick, and Isadora Veeren. 2025. Distributed Quantum Advantage for Local Problems. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing, STOC 2025, Prague, Czech Republic, June 23-27, 2025*. ACM. arXiv:2411.03240 https://arxiv.org/abs/2411.03240

[4] Alkida Balliu, Sebastian Brandt, Juho Hirvonen, Dennis Olivetti, Mikaël Rabie, and Jukka Suomela. 2021. Lower Bounds for Maximal Matchings and Maximal Independent Sets. *J. ACM* 68, 5 (2021), 39:1–39:30. doi:10.1145/3461458

[5] Alkida Balliu, Sebastian Brandt, Dennis Olivetti, and Jukka Suomela. 2020. How much does randomness help with locally checkable problems?. In *PODC '20: ACM Symposium on Principles of Distributed Computing, Virtual Event, Italy, August 3-7, 2020*, Yuval Emek and Christian Cachin (Eds.). ACM, 299–308. doi:10.1145/3382734.3405715

[6] Alkida Balliu, Mohsen Ghaffari, Fabian Kuhn, Augusto Modanese, Dennis Olivetti, Mikaël Rabie, Jukka Suomela, and Jara Uitto. 2025. Shared Randomness Helps with Local Distributed Problems. In *52nd International Colloquium on Automata, Languages, and Programming, ICALP 2025, July 8-11, 2025, Aarhus, Denmark*. arXiv:2407.05445 https://arxiv.org/abs/2407.05445

[7] Aviv Bick, Gillat Kol, and Rotem Oshman. 2022. Distributed Zero-Knowledge Proofs Over Networks. In *Proceedings of the 2022 ACM-SIAM Symposium on Discrete Algorithms, SODA 2022, Virtual Conference / Alexandria, VA, USA, January 9 - 12, 2022*, Joseph (Seffi) Naor and Niv Buchbinder (Eds.). SIAM, 2426–2458. doi:10.1137/1.9781611977073.97

[8] Nicolas Bousquet, Laurent Feuilloley, and Sébastien Zeitoun. 2024. Local Certification of Local Properties: Tight Bounds, Trade-Offs and New Parameters. In *41st International Symposium on Theoretical Aspects of Computer Science, STACS 2024, March 12-14, 2024, Clermont-Ferrand, France (LIPIcs, Vol. 289)*, Olaf Beyersdorff, Mamadou Moustapha Kanté, Orna Kupferman, and Daniel Lokshtanov (Eds.). Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 21:1–21:18. doi:10.4230/LIPICS.STACS.2024.21

[9] Sebastian Brandt, Orr Fischer, Juho Hirvonen, Barbara Keller, Tuomo Lempiäinen, Joel Rybicki, Jukka Suomela, and Jara Uitto. 2016. A lower bound for the distributed Lovász local lemma. In *Proceedings of the 48th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2016, Cambridge, MA, USA, June 18-21, 2016*, Daniel Wichs and Yishay Mansour (Eds.). ACM, 479–488. doi:10.1145/2897518.2897570

[10] Keren Censor-Hillel and Einav Huberman. 2025. Near-Optimal Resilient Labeling Schemes. In *28th International Conference on Principles of Distributed Systems (OPODIS 2024)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 35–1.

[11] Yi-Jun Chang, Tsvi Kopelowitz, and Seth Pettie. 2019. An Exponential Separation between Randomized and Deterministic Complexity in the LOCAL Model. *SIAM J. Comput.* 48, 1 (2019), 122–143. doi:10.1137/17M1117537

[12] Yi-Jun Chang, Gopinath Mishra, Hung Thuan Nguyen, Mingyang Yang, and Yu-Cheng Yeh. 2024. A Tight Lower Bound for 3-Coloring Grids in the Online-LOCAL Model. In *Proceedings of the 43rd ACM Symposium on Principles of Distributed Computing, PODC 2024, Nantes, France, June 17-21, 2024*, Ran Gelles, Dennis Olivetti, and Petr Kuznetsov (Eds.). ACM, 106–116. doi:10.1145/3662158.3662794

[13] Yi-Jun Chang and Seth Pettie. 2019. A Time Hierarchy Theorem for the LOCAL Model. *SIAM J. Comput.* 48, 1 (2019), 33–69. doi:10.1137/17M1157957

[14] Xavier Coiteux-Roy, Francesco D'Amore, Rishikesh Gajjala, Fabian Kuhn, François Le Gall, Henrik Lievonen, Augusto Modanese, Marc-Olivier Renou, Gustav Schmid, and Jukka Suomela. 2024. No Distributed Quantum Advantage for Approximate Graph Coloring. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing, STOC 2024, Vancouver, BC, Canada, June 24-28, 2024*, Bojan Mohar, Igor Shinkar, and Ryan O'Donnell (Eds.). ACM, 1901–1910. doi:10.1145/3618260.3649679

[15] Manuela Fischer and Mohsen Ghaffari. 2017. Sublogarithmic Distributed Algorithms for Lovász Local Lemma, and the Complexity Hierarchy. In *31st International Symposium on Distributed Computing, DISC 2017, October 16-20, 2017, Vienna, Austria (LIPIcs, Vol. 91)*, Andréa W. Richa (Ed.). Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 18:1–18:16. doi:10.4230/LIPICS.DISC.2017.18

[16] Orr Fischer, Rotem Oshman, and Dana Shamir. 2022. Explicit space-time tradeoffs for proof labeling schemes in graphs with small separators. In *25th International Conference on Principles of Distributed Systems (OPODIS 2021)*. Schloss Dagstuhl-Leibniz-Zentrum für Informatik.

[17] Mohsen Ghaffari, David G. Harris, and Fabian Kuhn. 2018. On Derandomizing Local Distributed Algorithms. In *59th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2018, Paris, France, October 7-9, 2018*, Mikkel Thorup (Ed.). IEEE Computer Society, 662–673. doi:10.1109/FOCS.2018.00069

[18] Mohsen Ghaffari, Fabian Kuhn, and Yannic Maus. 2017. On the complexity of local distributed graph problems. In *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2017, Montreal, QC, Canada, June 19-23, 2017*, Hamed Hatami, Pierre McKenzie, and Valerie King (Eds.). ACM, 784–797. doi:10.1145/3055399.3055471

[19] Mika Göös and Jukka Suomela. 2016. Locally checkable proofs in distributed computing. *Theory of Computing* 12 (2016), 1–33.

[20] Alex B. Grilo, Ami Paz, and Mor Perry. 2025. Distributed Non-Interactive Zero-Knowledge Proofs. *CoRR* abs/2502.05445 (2025). doi:10.48550/ARXIV.2502.07594 arXiv:2502.07594

[21] François Le Gall, Harumichi Nishimura, and Ansis Rosmanis. 2019. Quantum Advantage for the LOCAL Model in Distributed Computing. In *36th International Symposium on Theoretical Aspects of Computer Science, STACS 2019, March 13-16, 2019, Berlin, Germany (LIPIcs, Vol. 126)*, Rolf Niedermeier and Christophe Paul (Eds.). Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 49:1–49:14. doi:10.4230/LIPICS.STACS.2019.49

[22] Nathan Linial. 1992. Locality in Distributed Graph Algorithms. *SIAM J. Comput.* 21, 1 (1992), 193–201. doi:10.1137/0221015

[23] Augusto Modanese, Pedro Montealegre, and Martín Ríos-Wilson. 2025. Strong and Hiding Distributed Certification of k-Coloring. *CoRR* abs/2502.13854 (2025). doi:10.48550/ARXIV.2502.13854 arXiv:2502.13854

[24] Moni Naor and Larry J. Stockmeyer. 1995. What Can be Computed Locally? *SIAM J. Comput.* 24, 6 (1995), 1259–1277. doi:10.1137/S0097539793254571

[25] Václav Rozhon and Mohsen Ghaffari. 2020. Polylogarithmic-time deterministic network decomposition and distributed derandomization. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing, STOC 2020, Chicago, IL, USA, June 22-26, 2020*, Konstantin Makarychev, Yury Makarychev, Madhur Tulsiani, Gautam Kamath, and Julia Chuzhoy (Eds.). ACM, 350–363. doi:10.1145/3357713.3384298